RE501 Class 2: Probability Spaces, Conditional Probability and Random Variables

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In Class 1 (Section 1 and Section 7) we computed probability as a ratio of counts, $\Pr(E) = n(E)/n(\Omega)$. That rule needs a finite set of outcomes, all equally likely. In this class we build the general framework, called **probability theory**, which works without those two restrictions, and then use it to understand conditional probability, independence and random variables.

Chapter 3: The probability space

19 What is probability theory?

Probability theory is the mathematical study of random experiments. It describes a random experiment with three objects:

1. the **sample space** Ω: the set of all outcomes;
2. the **event space** $\sigma(\Omega)$: the collection of subsets of Ω to which we will assign a probability;
3. the **probability function** Pr: a rule that assigns a number to every event in $\sigma(\Omega)$.

The triple $(\Omega, \sigma(\Omega), \Pr)$ is called a **probability space**. Every question in probability is a question about some probability space, whether or not it is written down explicitly. The next three sections define each object.

20 Sample space

Sample space. The set of all possible outcomes of a random experiment is called sample space. We will denote the sample space by Ω (Greek capital "omega"). Each outcome is an *element* of Ω.

- Toss one coin: $\Omega = \{H, T\}$.
- Toss two coins: $\Omega = \{HH, HT, TH, TT\}$.
- Roll one die: $\Omega = \{1, 2, 3, 4, 5, 6\}$.
- Count the tosses needed to see the first Head: $\Omega = \{1, 2, 3, \dots\}$, an infinite set.

21 Event space: the σ-algebra

An **event** is a subset of Ω, for example "at least one Head" $= \{HH, HT, TH\}$. We say the event *occurs* if the outcome of the experiment is one of its elements.

We do not always need (or, for infinite sample spaces, are not always able) to assign a probability to *every* subset of Ω. Instead we pick a set of subsets of Ω, which is called an event space. We will denote the event space as $\sigma(\Omega)$.

If E is an event in the event space, the set relationship is as follows:

$$\begin{array}{ccc} 2^{\Omega} & \supseteq & \sigma(\Omega) \\ \downarrow & & \downarrow \\ \Omega & \supseteq & E \end{array}$$

However, when we ask some probabilistic questions, we often seek probability of composition of events - *In a two coin toss experiment, what is the probability of getting at least one head? What is the probability of getting no head at all?* corresponding to same event space, or *in the same two coin toss experiment what is the probability of getting the first one head? what is the probability of second one head? what is the probability of getting first one head or second one head?*

Clearly, we cannot pick any collection we like. If we can talk about the probability of some events, then logic forces us to also talk about some others. These requirements impose the following properties on the event space-

1. $\sigma(\Omega)$ should contain the sample space Ω,

$$\Omega \in \sigma(\Omega)$$

2. If event $E \in \sigma(\Omega)$ then $E^C = \Omega \setminus E \in \sigma(\Omega)$ - **the event space is closed under complementation**
3. For any number of events $E_1, E_2, \cdots \in \sigma(\Omega)$, their union $E_1 \cup E_2 \cup \cdots \in \sigma(\Omega)$ - **the event space is closed under countable unions**

Let us understand each criterion with the two coin toss, $\Omega = \{HH, HT, TH, TT\}$.

(i) The sample space is an event

Ω is the event "something happens". It always occurs, since every outcome lies in Ω. It is called the **sure event**. Any sensible event space must contain it.

(ii) Closed under complementation

If we can ask "did E occur?", then we have automatically also asked "did E not occur?". So if E is an event, E^c must be an event.

Example. If "at least one Head" $= \{HH, HT, TH\}$ is in $\sigma(\Omega)$, then its complement "no Head" $= \{TT\}$ must also be in $\sigma(\Omega)$.

Combining (i) and (ii): The null event $\emptyset = \Omega^c$ is also an element of event space. The null set is also called the **impossible event**.

(iii) Closed under countable unions

If E_1 and E_2 are events, "E_1 or E_2 occurred" must also be an event, i.e. $E_1 \cup E_2 \in \sigma(\Omega)$. Repeating this, any *finite* union $E_1 \cup E_2 \cup \cdots \cup E_n$ of events is an event. The criterion asks for a little more: even a *countably infinite* union $E_1 \cup E_2 \cup \ldots$ must be an event.

Example. If "both Heads" $= \{HH\}$ and "both Tails" $= \{TT\}$ are in $\sigma(\Omega)$, then "both coins show the same face" $= \{HH, TT\}$ must be in $\sigma(\Omega)$.

When Ω is finite, $\sigma(\Omega)$ is finite too, so any infinite union only repeats finitely many distinct sets; closure under finite unions is enough. The countable version matters for infinite sample spaces such as $\Omega = \{1, 2, 3, \dots\}$, where for example "the first Head comes on an even toss" $= \{2\} \cup \{4\} \cup \{6\} \cup \dots$ is an infinite union.

Using De Morgan's law (Class 1, Section 13), it can be shown that event spaces are also closed under intersections and set differences.

22 Worked example: two coin tosses

Sample space. $\Omega = \{HH, HT, TH, TT\}$, $n(\Omega) = 4$.

Power set. Since $n(\Omega) = 4$, the power set has $2^4 = 16$ elements:

Size of subset	Subsets	Count
0	$\emptyset$	1
1	$\{HH\}, \{HT\}, \{TH\}, \{TT\}$	4
2	$\{HH, HT\}, \{HH, TH\}, \{HH, TT\}, \{HT, TH\}, \{HT, TT\}, \{TH, TT\}$	6
3	$\{HH, HT, TH\}, \{HH, HT, TT\}, \{HH, TH, TT\}, \{HT, TH, TT\}$	4
4	$\{HH, HT, TH, TT\}$	1

The power set 2^Ω itself is a σ-algebra (check: it contains every subset, so it certainly contains Ω, all complements and all unions). It is the *largest* possible event space on Ω.

The smallest event space containing $\{HH\}$. Suppose we only care about the event "both Heads". What is the smallest σ-algebra $\sigma(\Omega)$ with $\{HH\} \in \sigma(\Omega)$? We add sets only when a criterion forces us to.

1. Start with what we want: $\{HH\}$.
2. Criterion (i): add $\Omega = \{HH, HT, TH, TT\}$.
3. Criterion (ii): add the complements. $\{HH\}^c = \{HT, TH, TT\}$ and $\Omega^c = \emptyset$.
4. Criterion (iii): check all unions of what we have. $\{HH\} \cup \{HT, TH, TT\} = \Omega$, and any union with $\emptyset$ or Ω gives back a set already present. Nothing new.
5. Check complements again: the complement of every set in the list is already in the list. Nothing new, so we stop.

$$\boxed{\sigma_{HH}(\Omega) = \{\, \emptyset,\ \{HH\},\ \{HT, TH, TT\},\ \Omega \,\}}$$

This event space has only 4 of the 16 subsets. It can answer exactly one question, "did both coins show Head, or not?". It cannot, for example, talk about "first coin shows Head" $= \{HH, HT\}$, because that set is not in $\sigma_{HH}(\Omega)$.

23 Probability function

Given a sample space Ω and a corresponding event space $\sigma(\Omega)$, the probability Pr is a function, $\Pr : \sigma(\Omega) \to [0, 1]$ that satisfies the following properties (the **axioms of probability**):

(P1) Probability of the sample space is 1: $\Pr(\Omega) = 1$.
(P2) Probability of mutually exclusive events $E_1, E_2, \cdots \in \sigma(\Omega)$ is the sum of the individual event probabilities,

$$\Pr(E_1 \cup E_2 \cup \dots) = \Pr(E_1) + \Pr(E_2) + \dots$$

The triple $(\Omega, \sigma(\Omega), \Pr)$ is called a **probability space**.

23.1 Some important properties

- **Probability of impossible event:** Since $\Omega = \Omega \cup \emptyset$ is a disjoint union, so $1 = 1 + \Pr(\emptyset)$ and we have $\Pr(\emptyset) = 0$
- **Monotonicity of probability** If $E_1 \subseteq E_2$ then $\Pr(E_1) \leq \Pr(E_2)$
- **Complement rule.** E and E^c are mutually exclusive and $E \cup E^c = \Omega$, so

$$\Pr(E^c) = 1 - \Pr(E).$$

The Class 1 ratio is a special case. (Class 1, Sections 1 and 7.) On a finite Ω with all outcomes equally likely, $\Pr(E) = n(E)/n(\Omega)$. Dividing by $n(\Omega)$ is exactly what makes $\Pr(\Omega) = 1$. The axioms, however, also allow biased experiments and experiments with infinitely many outcomes.

24 Simple and composite events

Simple (elementary) event. An event containing exactly one outcome, e.g. $\{HT\}$.

Composite (compound) event. An event containing more than one outcome. A composite event is a union of simple events, e.g. "at least one Head" $= \{HH\} \cup \{HT\} \cup \{TH\}$.

Example: two tosses of an unbiased coin. Take $\sigma(\Omega) = 2^\Omega$. "Unbiased" means all four simple events are equally likely. Since they are mutually exclusive and their union is Ω, axioms (P1) and (P2) give

$$\Pr(\{HH\}) + \Pr(\{HT\}) + \Pr(\{TH\}) + \Pr(\{TT\}) = \Pr(\Omega) = 1 \implies \text{each simple event has probability } \tfrac{1}{4}.$$

The probability of a composite event is then the sum over its simple events, by (P2):

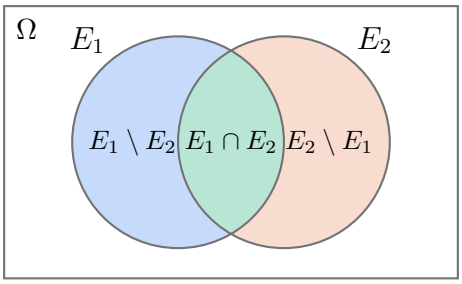
Event	Subset	Type	Probability
both Heads	$\{HH\}$	simple	$1/4$
no Head	$\{TT\}$	simple	$1/4$
first coin Head	$\{HH, HT\}$	composite	$1/4 + 1/4 = 1/2$
exactly one Head	$\{HT, TH\}$	composite	$1/2$
at least one Head	$\{HH, HT, TH\}$	composite	$3/4$
sure event	Ω	composite	1

Check: $\Pr(\text{at least one Head}) = 1 - \Pr(\text{no Head}) = 1 - 1/4 = 3/4$, by the complement rule.

25 Probability of a union

We show

$$\boxed{\Pr(E_1 \cup E_2) = \Pr(E_1) + \Pr(E_2) - \Pr(E_1 \cap E_2)}$$



For finite outcomes, this can be shown easily. From Class 1, Section 11, $n(E_1 \cup E_2) = n(E_1) + n(E_2) - n(E_1 \cap E_2)$. Divide both sides by $n(\Omega)$:

$$\frac{n(E_1 \cup E_2)}{n(\Omega)} = \frac{n(E_1)}{n(\Omega)} + \frac{n(E_2)}{n(\Omega)} - \frac{n(E_1 \cap E_2)}{n(\Omega)} \implies \Pr(E_1 \cup E_2) = \Pr(E_1) + \Pr(E_2) - \Pr(E_1 \cap E_2).$$

But for general cases, this can be shown from the probability axioms. Here the same three-piece split works, with Pr in place of n, because the only thing we used about n was additivity over mutually exclusive sets.

Step 1. $E_1 \cup E_2 = (E_1 \setminus E_2) \cup (E_1 \cap E_2) \cup (E_2 \setminus E_1)$, three mutually exclusive events. By (P2):

$$\Pr(E_1 \cup E_2) = \Pr(E_1 \setminus E_2) + \Pr(E_1 \cap E_2) + \Pr(E_2 \setminus E_1)$$

Step 2. $E_1 = (E_1 \setminus E_2) \cup (E_1 \cap E_2)$ and $E_2 = (E_2 \setminus E_1) \cup (E_1 \cap E_2)$, each a disjoint union. By (P2):

$$\Pr(E_1 \setminus E_2) = \Pr(E_1) - \Pr(E_1 \cap E_2), \qquad \Pr(E_2 \setminus E_1) = \Pr(E_2) - \Pr(E_1 \cap E_2)$$

Step 3. Substitute into Step 1:

$$\Pr(E_1 \cup E_2) = \big(\Pr(E_1) - \Pr(E_1 \cap E_2)\big) + \Pr(E_1 \cap E_2) + \big(\Pr(E_2) - \Pr(E_1 \cap E_2)\big) = \Pr(E_1) + \Pr(E_2) - \Pr(E_1 \cap E_2).$$

(All the sets used are in $\sigma(\Omega)$, because a σ-algebra is closed under intersection and difference.)

Example. E_1 = "first coin Head" = $\{HH, HT\}$, E_2 = "second coin Head" = $\{HH, TH\}$. Then $\Pr(E_1 \cup E_2) = \frac{1}{2} + \frac{1}{2} - \frac{1}{4} = \frac{3}{4}$, which matches $\Pr(\{HH, HT, TH\}) = 3/4$.

Chapter 4: Conditional probability and independence

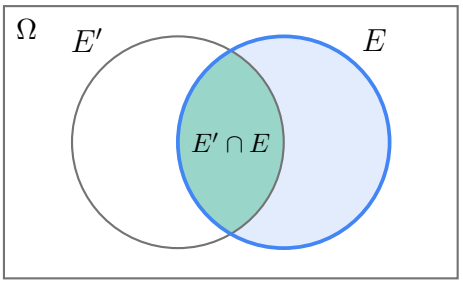
26 Conditional probability

Suppose we are told that event E has occurred, but not which outcome. How likely is E' now?

Finite sets, equally likely outcomes

Once we know E occurred, every outcome outside E is ruled out. E **becomes the new sample space.** Inside it, E' can only occur through the outcomes in $E' \cap E$. So

$$\Pr(E' \mid E) = \frac{n(E' \cap E)}{n(E)} \qquad \text{read: "probability of } E' \text{ given } E\text{".}$$



Given E: only the region inside E is possible.

$$\Pr(E' \mid E) = \frac{n(E' \cap E)}{n(E)}$$

Dividing numerator and denominator by $n(\Omega)$:

$$\Pr(E' \mid E) = \frac{n(E' \cap E)/n(\Omega)}{n(E)/n(\Omega)} = \frac{\Pr(E' \cap E)}{\Pr(E)}.$$

General definition

The last expression makes sense in any probability space, so we take it as the definition:

Conditional probability. For events $E', E \in \sigma(\Omega)$ with $\Pr(E) > 0$,

$$\Pr(E' \mid E) = \frac{\Pr(E' \cap E)}{\Pr(E)}.$$

Example. Two tosses of an unbiased coin. E' = "both Heads" $= \{HH\}$, E = "at least one Head" $= \{HH, HT, TH\}$. Then $E' \cap E = \{HH\}$ and

$$\Pr(E' \mid E) = \frac{n(\{HH\})}{n(\{HH, HT, TH\})} = \frac{1}{3},$$

whereas without the information $\Pr(E') = 1/4$. Knowing E changed the probability of E'.

Multiplication rule

Rearranging the definition, and writing it also with the roles of E' and E swapped (note $E' \cap E = E \cap E'$):

$$\boxed{\Pr(E \cap E') = \Pr(E \mid E')\ \Pr(E') = \Pr(E' \mid E)\ \Pr(E)}$$

This is the form we use most: the probability that *both* happen is the probability that one happens times the probability that the other happens given the first.

27 Independence

Independent events. E and E' are **independent** if knowing that E' occurred does not change the probability of E:

$$\Pr(E \mid E') = \Pr(E).$$

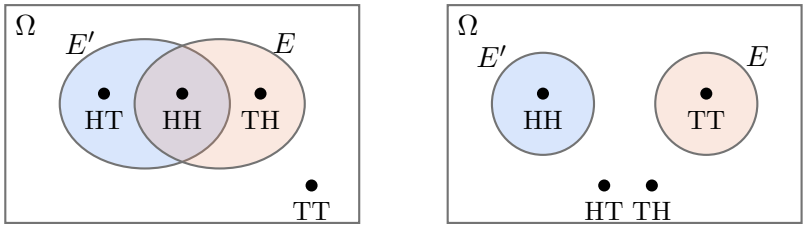
Putting this into the multiplication rule gives the equivalent (and symmetric) form:

$$\boxed{E, E' \text{ independent} \iff \Pr(E' \cap E) = \Pr(E')\ \Pr(E)}$$

Example. E' = "first coin Head" $= \{HH, HT\}$, E = "second coin Head" $= \{HH, TH\}$. $\Pr(E' \cap E) = \Pr(\{HH\}) = 1/4 = \frac{1}{2} \times \frac{1}{2} = \Pr(E')\Pr(E)$. Independent: what the first coin did tells us nothing about the second.

27.1 Mutually exclusive is not the same as independent

These two ideas are often confused, but they are almost opposites. E' and E are **mutually exclusive** events when $E' \cap E = \emptyset$. In that case $\Pr(E' \cap E) = 0$, so they are independent only if $\Pr(E')\Pr(E) = 0$, i.e. only if $\Pr(E') = 0$ or $\Pr(E) = 0$. **Mutually exclusive events with non-zero probabilities are always dependent**.



E' = first Head, E = second Head overlap, **independent**

E' = both Heads, E = both Tails mutually exclusive, **dependent**

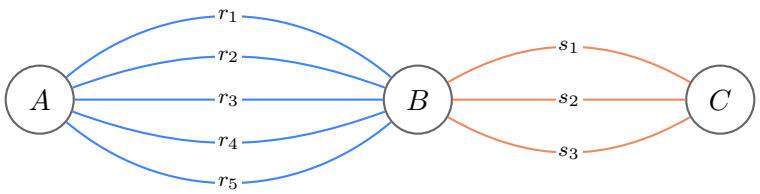
Example of mutually exclusive events and dependent events: Take $E' = \{HH\}$ and $E = \{TT\}$, each with probability 1/4. If we learn that E occurred (both Tails), then E' (both Heads) is now impossible:

$$\Pr(E' \mid E) = \frac{\Pr(\emptyset)}{\Pr(E)} = 0 \neq \tfrac{1}{4} = \Pr(E').$$

	Mutually exclusive	Independent
Definition	$E' \cap E = \emptyset$	$\Pr(E' \cap E) = \Pr(E')\Pr(E)$
Property of	the two sets only	the probability function
Union rule	$\Pr(E' \cup E) = \Pr(E') + \Pr(E)$	$\Pr(E' \cup E) = \Pr(E') + \Pr(E) - \Pr(E')\Pr(E)$
Intersection	$\Pr(E' \cap E) = 0$	$\Pr(E' \cap E) = \Pr(E')\Pr(E)$
Meaning	"cannot happen together"	"one tells nothing about the other"

28 Independence in daily life: counting routes

Question. From bus depot A to depot B there are 5 routes. From B to depot C there are 3 routes. How many ways are there to travel from A to C via B?



Answer. $5 \times 3 = 15$. Everyone "knows" this, but why multiplication? Each journey is a pair (route $A \to B$, route $B \to C$):

$$\Omega = \{r_1s_1, r_1s_2, r_1s_3, r_2s_1, \ldots, r_5s_3\}, \qquad n(\Omega) = 5 \times 3 = 15.$$

The count is 15 only because **every** $B \to C$ route can follow **every** $A \to B$ route: the choice of the second route does not depend on the first. That is independence.

In the language of probability. Suppose a traveller picks the $A \to B$ route at random (each with probability 1/5) and, independently, the $B \to C$ route at random (each 1/3). Let E_i = "first route on r_i" and E'_j = "second route on s_j". By independence,

$$\Pr\left(\{r_is_j\}\right) = \Pr(E_i \cap E'_j) = \Pr(E_i)\ \Pr(E'_j) = \frac{1}{5} \times \frac{1}{3} = \frac{1}{15}.$$

So there are 15 equally likely journeys. The multiplication rule of counting is the product rule for independent events in disguise.

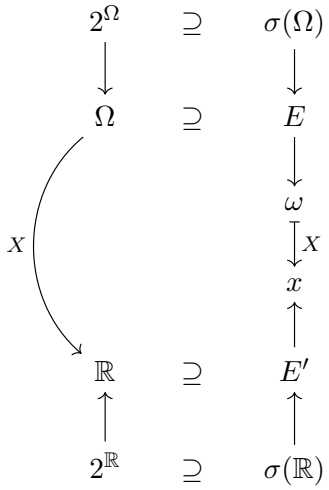
When independence fails. Suppose a bus on route r_1 always reaches B after the last bus on route s_3 has left. Now the second route *depends* on the first: after r_1 only 2 continuations are possible. The number of journeys is $4 \times 3 + 1 \times 2 = 14$, not 15, and simple multiplication gives the wrong answer.

We use independence like this all the time, often without noticing. The framework of this chapter lets us see exactly what we are assuming when we multiply, and when that assumption is wrong.

Chapter 5: Random variables

29 Random variable as a measurable function

The sample space and event space of probability theory contain abstract objects. Working with these abstract objects is cumbersome. It is always better to work with real numbers. So we define a function $X : \Omega \to \mathbb{R}$, i.e. a function from abstract sample space to real number line. What are the required properties of the function?



Firstly, to preserve the structure of the event space, we have to make an analogous measurable space $(\mathbb{R}, \sigma(\mathbb{R}))$, which will be just like the space $(\Omega, \sigma(\Omega))$. The function X need not be one-to-one (different outcomes may give the same value; in the example below both HT and TH give 1), nor onto (some real numbers may never be produced). But for every event $E' \in \sigma(\mathbb{R})$, the set of outcomes that X sends into E' must be an event in $\sigma(\Omega)$, so that Pr can measure it.

Formally this is written as follows: for a given probability space $(\Omega, \sigma(\Omega), \Pr)$ and measurable space $(\mathbb{R}, \sigma(\mathbb{R}))$, the function $X : \Omega \to \mathbb{R}$ must satisfy, for any event $E' \in \sigma(\mathbb{R})$,

$$X^{-1}(E') = \{\omega \in \Omega \mid X(\omega) \in E'\} \in \sigma(\Omega).$$

The function X is called a **Random Variable**. The probability of the values that X can produce is determined by the probability of the preimage event, i.e. for any $E' \in \sigma(\mathbb{R})$,

$$\Pr_X(E') = \Pr\left(X^{-1}(E')\right) = \Pr\left(\{\omega \in \Omega \mid X(\omega) \in E'\}\right).$$

In particular, for a single value x we write $\Pr_X(x)$ for

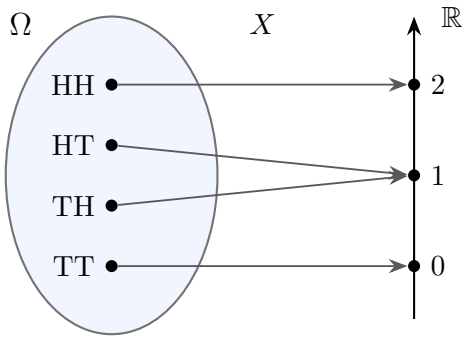
$$\Pr_X(x) = \Pr_X(\{x\}) = \Pr(X = x) = \Pr\big(\{\omega \in \Omega \mid X(\omega) = x\}\big).$$

Since the preimage is an event of the probability space $(\Omega, \sigma(\Omega), \Pr)$, its probability is well defined.

It is important to distinguish two functions here. $\Pr_X$ 'eats up' an event $E' \in \sigma(\mathbb{R})$, i.e. a *set* of real numbers, and produces a number in $[0, 1]$. It satisfies both axioms: $\Pr_X(\mathbb{R}) = \Pr(\Omega) = 1$, and preimages of mutually exclusive sets are mutually exclusive, so the probabilities add. Hence $(\mathbb{R}, \sigma(\mathbb{R}), \Pr_X)$ is itself a probability space, called the **distribution** of X. The function $x \mapsto \Pr_X(x) = \Pr(X = x)$, on the other hand, eats up a *single* real number x and produces a number in $[0, 1]$. It is called the **probability mass function** of X.

30 Example: number of Heads in two tosses

Two tosses of an unbiased coin, $\sigma(\Omega) = 2^\Omega$. Let X = number of Heads.



Value x	Preimage $X^{-1}(x)$	Type of event	$\Pr_X(x)$
0	$\{TT\}$	simple	1/4
1	$\{HT, TH\}$	composite	1/2
2	$\{HH\}$	simple	1/4

The probabilities add to 1, as they must, since the three preimages are mutually exclusive and their union is Ω. This table is called the **probability distribution** of X.

Chapter 6: Quantification of distribution

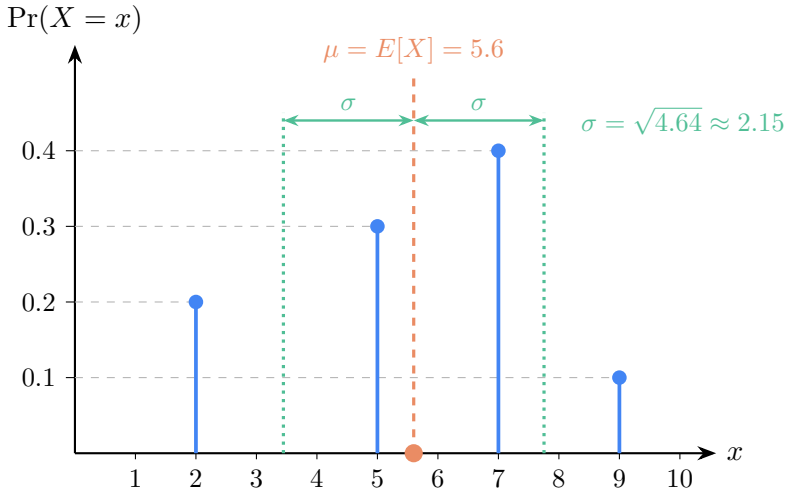
31 Probability distribution of a discrete random variable

Once we have **Random Variable** (R.V.), we can use this to ask different questions like, what is the average value of random variable, or if we perform a **random experiment**, how likely the resulting **Random Variable** that we get will be away from a given number etc.

Consider the **probability distribution** of **random variable** X which produces some discrete values-

Value x	$\Pr(X = x)$
2	0.2
5	0.3
7	0.4
9	0.1

If we plot in the x-y plane, with values of X along the x-axis and the corresponding probability $\Pr_X(x)$ along the y-axis, we get something like the following-



32 Mean (expectation) of a random variable

We can calculate the mean of the X as follows:

$$\bar{X} = E[X] = \sum_i x_i \Pr_X(x_i)$$

From probability distribution the mean can be calculated as follows-

Value x	$\Pr(X = x)$	$x \Pr_X(x)$
2	0.2	0.4
5	0.3	1.5
7	0.4	2.8
9	0.1	0.9
sum		5.6

The mean is then, 5.6. The mean is *always* a number.

To understand this intuitively, assume taking a large number of repetitions of the random experiment. The average count of the different values in such a case can be estimated as-

Value x	$\Pr(X = x)$	repetition	sum of x
2	0.2	20	$2 \times 20 = 40$
5	0.3	30	$5 \times 30 = 150$
7	0.4	40	$7 \times 40 = 280$
9	0.1	10	$9 \times 10 = 90$
sum			560

Since the sum is $\sum_i x_i(N \Pr_X(x_i))$ and total repetition of the experiment is N, the average x is,

$$\frac{1}{N}\sum_i x_i(N \Pr_X(x_i)) = \sum_i x_i \Pr_X(x_i)$$

Mean gives the *expected value of a random variable.*

33 Variance and standard deviation

Another measure of the R.V. X is how far, typically, a number produced in one random experiment lies from the mean value. This is measured by the **variance**, denoted by σ_X^2, defined as-

$$\sigma_X^2 = \sum_i (x_i - \bar{X})^2 \Pr_X(x_i)$$

The non-negative square root of the variance, σ_X, is called the **standard deviation**.

In the above example we can calculate the variance as follows-

- First calculate mean, as shown above. This gives $\bar{X} = 5.6$
- Use the following table-

Value x	$\Pr(X = x)$	$x - \bar{X}$	$(x - \bar{X})^2$	$(x - \bar{X})^2 \Pr_X(x)$
2	0.2	−3.6	12.96	2.592
5	0.3	−0.6	0.36	0.108
7	0.4	1.4	1.96	0.784
9	0.1	3.4	11.56	1.156
sum				4.64

This gives the variance $\sigma_X^2 = 4.64$ and standard deviation $\sigma_X \approx 2.154$

Standard deviation gives an idea of how spread the values of R.V. X are.

The variance is always a non-negative real number. By definition, the standard deviation is also a non-negative real number.

34 n-th moment of a random variable

The above idea can be generalized to define the n-th order **central moment**:

$$\mu_n(X) = \sum_i (x_i - \bar{X})^n \Pr_X(x_i)$$

The *central* here is meaning the mean value $\bar{X}$. Correspondingly, another way to measure the R.V. moments are using n-th order **raw moment**-

$$E[X^n] = \langle X^n \rangle = \sum_i x_i^n \Pr_X(x_i)$$

34.1 First central moment is always zero

$$\begin{aligned}
\mu_1(X) &= \sum_i (x_i - \bar{X}) \Pr_X(x_i) \\
&= \Big[\sum_i x_i \Pr_X(x_i)\Big] - \bar{X} \sum_i \Pr_X(x_i) \\
&= \bar{X} - \bar{X} = 0
\end{aligned}$$

34.2 Second central moment (variance) in terms of raw moments

$$\begin{aligned}
\sigma_X^2 = \mu_2(X) &= \sum_i (x_i - \bar{X})^2 \Pr_X(x_i) \\
&= \sum_i (x_i^2 - 2\bar{X}x_i + \bar{X}^2) \Pr_X(x_i) \\
&= \Big[\sum_i x_i^2 \Pr_X(x_i)\Big] - 2\bar{X}\sum_i x_i \Pr_X(x_i) + \bar{X}^2 \sum_i \Pr_X(x_i) \\
&= E[X^2] - \bar{X}^2
\end{aligned}$$

This can be useful to calculate the variance for a probability distribution at one go.

For the above example,

Value x	$\Pr(X = x)$	x^2	$x \Pr_X(x)$	$x^2 \Pr_X(x)$
2	0.2	4	0.4	0.8
5	0.3	25	1.5	7.5
7	0.4	49	2.8	19.6
9	0.1	81	0.9	8.1
sum			5.6	36

The variance is, $\sigma_X^2 = \langle X^2 \rangle - \bar{X}^2 = 36 - 5.6^2 = 4.64$ as expected.